Relaxation Processes

For a particle moving through a background of scatterers, we have two main types of relaxation processes:

Energy loss and momentum loss.

Let's look at both of them.

Energy loss

For an initially stationary target, final velocity in lab frame is:

$$v_L' = \left(\frac{m_1 v_1}{m_1+m_2} + \frac{m_2 v_1}{m_1+m_2} \cos\chi_c , \frac{m_2 v_1}{m_1+m_2} \sin\chi_c \right)$$

This gives us the final K.E as: $K' = \frac{1}{2} m_1 v_L'^2 = \frac{1}{2} m_1 v_1^2 \left\{ 1 - \frac{2 m_1 m_2}{(m_1+m_2)^2} - 2\sin^2\frac{\chi_c}{2} \right\}$

There is visibly a loss in kinetic energy of this particle:

$$\Delta K = K - K' = \frac{1}{2} m_1 v_2 \frac{4 m_1 m_2}{(m_1+m_2)^2} - \sin^2\frac{\chi_c}{2} = \frac{1}{2} m_1 v_1^2 \frac{4 m_1 m_2}{(m_1+m_2)^2} - \frac{1}{\left(\frac{b}{b_{90}}\right)^2 + 1}$$

Notice how $\Delta K \to 0$ as $\chi_c \to 0$

For small angles ($\chi_c \ll 1$, $b/b_{90} \gg 1$)

$$\Delta K = \left(\frac{1}{2} m_1 v_1^2 \right) \frac{4 m_1 m_2}{m_1+m_2} - \left(\frac{b_{90}}{b} \right)^2$$

Now that we have the energy lost per collision, we can look at how much energy a projectile loses p.u length.

This will be: $n_2 \, dl \, 2\pi b \, db \, \frac{1}{2} m_1 v_1^2 \frac{4 m_1 m_2}{(m_1+m_2)^2} - \left(\frac{b_{90}}{b} \right)^2$

Running the required integral gives us:

$$\frac{dK}{dl} = K n_2 \frac{m_1 m_2}{(m_1+m_2)^2} 8\pi b_{90}^2 \left[\ln|b| \right]_{b_{min}}^{b_{max}}$$

$\longrightarrow$ The problem with this result is that the limits $b \to 0$, $b \to \infty$ diverge logarithmically.

This analysis is not accurate because we are using the small angle approximation and we are ignoring the plasma shielding.

Plasma shielding will limit the range of the coulomb force to the debye length.

The small angle appx. breaks down at $b \sim b_{90}$. Thus one solution to solve both of these problems is to cut the integral off at: $b_{min} = b_{90}$, $b_{max} = \lambda_D$

This gives us: $\frac{dK}{dl} = K n_2 \frac{m_1 m_2}{(m_1+m_2)^2} 8\pi b_{90}^2 \ln|\Lambda| \longrightarrow \Lambda = \frac{\lambda_D}{b_{90}} = \left(\frac{\epsilon_0 T_e}{n e^2} \right)^{1/2} \Big/ \left(\frac{q_1 q_2}{4\pi\epsilon_0 m_r u^2} \right)$

$\ln\Lambda$ is known as the Coulomb logarithm. This cut-off also comes with the advantage that due to its logarithmic nature, we do not have to worry about getting the most precise values.

A commonly used term is collision frequency. The energy loss collision frequency will be given by:

$$\nu_K \equiv v_1 \frac{1}{K}\frac{dK}{dL} = n_2 v_1 \frac{m_1 m_2}{(m_1+m_2)^2} 8\pi b_{90}^2 \ln|\Lambda| = n_2 \frac{q_1^2 q_2^2}{(4\pi\epsilon_0)^2} \frac{8\pi}{m_1 m_2 v_1^3} \ln\Lambda$$

We can also find the effective energy-loss cross-section: $\sigma_K \equiv \frac{1}{K n_2}\frac{dK}{dL} = \frac{q_1^2 q_2^2}{(4\pi\epsilon_0)^2}\frac{8\pi}{m_1 m_2 v_1^4}\ln\Lambda = \frac{\nu_K}{n_2 v_1}$

↳ Instead of giving us details about scattering angle,

This cross-section tells us about the energy losses.

Momentum Loss

Loss of momentum in x-dir^n after 1 collision:

$$\Delta p_x = m_1(v_1 - v_{1x}') = p_x \frac{m_2}{m_1+m_2}(1-\cos\chi_c) \approx p_x \frac{m_2}{m_1+m_2}\frac{\chi_c^2}{2} = p_x \frac{m_2}{m_1+m_2}\frac{2b_{90}^2}{b^2}$$

We can do what we did for the Energy loss to get: $\frac{dp}{dL} = n_2 p \frac{m_2}{m_1+m_2} 4\pi b_{90}^2 \ln\Lambda$

Keep in mind that: $\frac{dp}{dt} = v\frac{dp}{dL} = n_2 v^2 \frac{m_1 m_2}{m_1+m_2} 4\pi b_{90}^2 \ln\Lambda$

Similar to the energy loss, we can calculate the collision frequency: $\nu_p = n_2 \frac{q_1^2 q_2^2}{(4\pi\epsilon_0)^2}\frac{4\pi(m_1+m_2)}{m_2 m_1^2 v_1^3}\ln\Lambda$

An Interesting fact about energy and momentum loss is their ratio:

$$\frac{\text{Energy loss } \nu_K}{\text{momentum loss } \nu_p} = \frac{2}{m_1 m_2} \Big/ \frac{m_1+m_2}{m_2 m_1^2} = \frac{2m_1}{m_1+m_2} \longrightarrow \approx 2 \text{ if } m_1 \gg m_2;\ = 1 \text{ if } m_1 = m_2;\ \ll 1 \text{ if } m_1 \ll m_2$$

This way, for electrons, momentum loss is much greater than energy loss. Therefore it is mostly the angle of velocity that scatters.

Momentum scattering time is often called the 90° scattering time or time to diffuse through a 90° angle.

For the $m_1 \ll m_2$ case, all that we really have to worry about is the scattering angle: $\chi_L \approx \chi_c = 2b_{90}/b$

The momentum deviations are equally likely in all directions. This means that the mean angle of deviation in the length L is 0. However, the mean squared deviation angle is not:

$$\overline{\Delta\alpha^2} = n_2 L \int^{b_{max}} \chi^2 2\pi b\,db = L n_2 8\pi b_{90}^2 \ln\Lambda$$

bmin